SOLID WASTE MANAGEMENT IN BENGALURU WITH REFERENCE TO BENGALURU EAST ZONE

Social Immersion Report submitted in partial fulfillment of the requirements for the award of the Degree of

MASTER OF BUSINESS ADMINISTRATION

OF

BENGALURU CENTRAL UNIVERSITY

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BENGALURU CITY UNIVERSITY
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BE BOUNDLESS

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2022-2024

DECLARATION BY THE STUDENT

I hereby declare that the **"SOLID WASTE MANAGEMENT IN BENGALURU WITH REFERENCE TO BENGALURU EAST ZONE"** is the result of the work carried out by me under the guidance of **DR. NIRMALA M** in partial fulfillment for the award of a Master's Degree in Business Administration by Bengaluru Central University.

I also declare that this report is the outcome of my own efforts and that it has not been submitted to any other University or Institute for the award of any other Degree/Diploma/Certificate. I also declare that the business immersion report is not presented in any forum earlier than this.

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GUIDE CERTIFICATE

This is to certify that the Business Plan Report **"SOLID WASTE MANAGEMENT IN BENGALURU WITH REFERENCE TO BENGALURU EAST ZONE"** Submitted by **PRAFUL (P18UZ22M015036)** to Bengaluru City University, Bengaluru for the award of Degree of Master of Business Administration is a record of work carried out by him/her under my guidance.

Place: Bangalore

Date: Signature

CERTIFICATE OF ORIGINALITY

This is to certify that the Business Plan report entitled **"SOLID WASTE MANAGEMENT IN BENGALURU WITH REFERENCE TO BENGALURU EAST ZONE"** is an original work of **Mr. PRAFUL** bearing University Register Number **P18UZ22M015036** and is being submitted in partial fulfillment for the award of the Master's Degree in Business Administration of Bengaluru City University. The report has not been submitted earlier to any University /institution to fulfill the requirement of any course of study. **Mr. PRAFUL** is guided by **DR. NIRMALA M** who is the Faculty Guide as per the regulations of Bengaluru City University.

SIGNATURE AND SEAL OF DIRECTOR / PRINCIPAL /HOD

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CHAPTER-01

1. INTRODUCTION

1.1 Background of the study

The Bruhat Bengaluru Mahanagara Palike (BBMP) is an administrative of various departments. Solid Waste Management is among departments in BBMP; it is the responsibility of BBMP to maintain the SWM department.

It covers an area of 713sq km and the housing population is approximately 1.3 corers and it is among fastest growing cities in India. Everyday Bengaluru produces lots of solid waste it is around 5000 tons per day and rises to 6000tons by 2029.

BBMP SWM department operates in 8 zones 27 divisions and 198 wards each zones is managed by Officers, Private Agencies, Engineers, Employees, Pourakarmikas and marshals etc.

For disposal BBMP has some waste processing plant it is also known as waste treatment plant where it can handle different types of waste such as solid waste industrial waste and recyclable materials and other waste and BBMP also has some bio-methanation Apart from this it has sanitary landfilling area for disposal of waste that plays an important role of in waste management.

In East Zone, it consists of 25 wards; Domlur (ward 112) is one of them. The dumping yard is near Domlur flyover, waste segregation system carries the use of Auto tippers , pushcarts with primary segregation being done from house- to- house visit .

Domlur ward 112, there are approximately 750 houses, for Segregation there are 8 small auto tippers and 2 big tippers and there are many CCTV cameras installed, Domlur ward association members and ward officers regularly monitor and take care for peoples.

Associations are working to solve problems, by arranging monthly meetings like ward welfare association meeting and residential welfare associations (RWAs).

1.2 Need of the study:

- I previously participated as volunteer in Brand Bengaluru organized by BBMP. Topic was solid waste management, got peculiar to study further on solid waste management.
- People come from different states, districts; villages reside in Bengaluru for work. While going to work or coming from office, they throw garbage on road, instead of auto tippers/ dustbin collector I observed this, studying would help to gain information root cause and identify solutions
- To gain knowledge about types of waste generated, segregation of waste disposal, what challenges they faced, transportation process, types of technology used, and types of agencies that participated in the SMW
- The need of the study is provide suggestions and responsibilities to people. So that people's will know their responsibilities to manually segregate wet and dry waste currently people are mixing all their waste in one bag and not following guidelines by using colour bins, which becomes very difficult to segregate.
- Another need of Study is how to give information to house owners and tenant families .The tenants and house owner shall register in nearest ward. The Owners of 4 and 5 floors, only owner is registered in the wards and tenant's families are not registered. This causing only waste received from owner family, tenants wastage is not received Around 750 houses are supposed to give garbage but only 250 houses are giving garbage remaining houses are not giving.
- Investigating why people are doing illegal dumping despite the availability of guidelines, because individuals are not doing their responsibility BBMP and NGO are unable to resolve this illegal dumping.

1.3 Social relevance of the topic:

There is a significant potential in project to study on management of solid waste with social significance, and also provides very important foundation for transformation in several forms.

- **Public Health and Well-being:** Good practices of disposal of waste decreases the risk of many dangerous and infectious diseases, which are mainly because of improper waste management. For example dengue fever, malaria, viral fever, etc. and respiratory infections, are because of unmanaged waste practices. Therefore the effective practices of waste segregation and management are required for the safety of public health and improve the standard of life for the communities in Bengaluru

- **Environmental Protection:** Adherence to best practices is crucial for the conservation of natural environment and for encountering the bad impacts of pollution for environment. The research helps to provide ways to better solid waste disposal strategies, these help for conservation of environment soil fertility, quality of drinking water, water for crops, and conservation of biodiversity.

- **Livelihood Opportunities:** This improvement in practices of managing waste will create the opportunity for employment and improve the economy of local community. The new ideas targeted at segregation of waste, recycling of waste, conversion of energy from waste will foster the livelihood of low income populations, like the garbage pickers, non-regular waste collectors and hence improving the economy.

- **Urban Sustainability:** Better ways of waste management or sustainable ways are very significant for the long term sustainability of the metropolitan cities like Bengaluru. The findings is to provide the ideas for developing policies, strategies for community engagement and building infrastructure strategies for building resilient and sustainable cities which are equipped to sustain the requirements and foster the needs of present generations and future generations.

Overall, solid waste management in Bengaluru is highly important and relevant to the society this helps to improve public health, conservation and prevention of the environment and urban sustainability

CHAPTER-02

2. Research Methodology

2.1 Objective of the study

This project seeks to educate the population about the effects of health and safety in BBMP East Zone Domlur Ward 112 caused by dumping of wastes within the interior part of some areas. By using qualitative tools and techniques in the study, such as site identification and analysis, public education on waterborne diseases and contaminated soils, and global SWM practices, the study aims at instilling motivation in the residents to adhere to regulations and help in the provision of good SWM strategies.

2.2 Statement of the Problem

The current waste management in Bengaluru East zone Domlur ward (112), it faces various challenges and they are:

- when people are going and coming in the early and late night and early morning they throw their garbage on the road near the dump yard and vehicles are going on that falling down

- The cities should be planned like developed countries but due to the population increase, many people are coming and going from other states, cities, and Villages, increasing population rapidly in the Bengaluru city the system isn't able to handle huge waste generated.

- People carry their waste in cars and put it outside the yard. Some people cover their noses as they leave the yard. Based on primary data, people claim to be paying, for example, x tax, but the amount of tax they are actually paying is 10 times higher than what is required to remove their garbage.

- Individuals are failing to fulfil their obligations. Authorities from NGOs and the BBMP are unable to stop illicit dumping.

- People are flushing the solid waste and also the sanitary pads into the flush, which is causing the drainage chambers to be blocked. This is very difficult to clean drainage chambers.

- The citizen's responsibility to manually segregate of wet and dry waste. People putting broken glasses, cloth, food items, and plastic in the same waste, segregation becomes very difficult.

- The garbage is dumped under the soil by digging and making big holes/waste dumping places, here already the water is contaminated, where the waste is dumped. People are getting diseases, because of drinking contaminated water near the dump yards.

- Their garbage is dumped due to chemical contamination in to soil crops are not growing and also causes disease to people due to Le hang happened.

- The permission to build the house in a 20 x 30 sq. ft. site is Ground +1, Ground +2 but the people are constructing 7 floors, which is causing the population increase, and hence the waste is also increasing. Hence when constructing houses BBMP doesn't permit to house owner but they give permission.

- In rainy seasons due to illegal Dumping 5kg of waste becomes 20 kg and there are no sufficient vehicles to carry this waste causing diseases such as dengue and making people are given complaint to BBMP to take care of illegal dumping.

- The road in front of the dump yard is not well done, which is not making the water flow to the drainage. People are taking cars on the dumped waste.

2.3 Methodology Employed:

- In this study, I used different types of methods to collect Information, first I took information from the BBMP website; there I got data about types of waste, segregation of waste, officers, and structure of the organization, and others.

- Secondly, I studied newspapers Such as the Hindu there I got only some information.

- After that, I previously participated as a volunteer in brand Bengaluru which is primary data that helped me a lot where I met officers and all teams explaining their thoughts problems, and suggestions to the deputy chief minister of Karnataka

- For more primary data I needed so, I visited and interviewed Domlur ward residential association officers.

- Lastly, for primary data Segregation, I interviewed officers, workers, drivers, and Pourakarmikas of Domlur yard all were very helpful, and they gave a lot of information.

- I also went to the place where people dump trash (waste) illegally and saw what happened there.

- For secondary data, I used research papers.

- All this information helped me to understand the waste situation in the east zone of Domlur (Ward 112).

CHAPTER-03

3. Brief Profile

3.1 HISTORY OF ORGANIZATION:-

Figure 1 **BBMP Head Office**

3.1.1 The Initial Period 1862-1881

- Bengaluru began its form of municipal governance in 1862 with the formation of the first Municipal Board on the 27th of March under the Improvement of Towns Act of 1850. This was organised by nine notable persons of the old city during that time. Initially, two distinct Municipal Boards were established, one for Bangalore City and Other for the Cantonment area.

- By **1881**, British government decided to retrograde Bangalore city under the control of Wodeyar kings. Meanwhile, the Bangalore City Municipality and the Civil, Military Station Municipality (Cantonment) must have been established and functioning separately. These boards seen to plan and coordinate the growth of the city in Bengaluru.

- **The Infrastructure and Sanitation:** These were Major challenges that early municipal governance faced and the need to improve physical infrastructure and public hygiene. Operations were taken to inflate the drainage and road infrastructure, and general health amenities within the city.

- **Income tax:** As a source of revenue for implementing the development of urban areas, property tax was imposed. This has helped to provide a steady stream of revenue for the municipal boards required for civil development projects.

- **Police and Local Improvement Powers:** The municipal boards secured powers from the police and local improvement body. This helped to maintain law and order, and undertaking urban development projects.

- **Legalization and Formalization:** The British government again handed over the Bangalore city to the Wodeyars in the year of 1881. At this time legal Bangalore City Municipality and Civil and military station Municipality (Cantonment) was formed. This formal recognition marked a clear evolution of a system of more defined and organized urban administration.

- **Independent Operations:** The two municipalities were independent, separate organizations with complete authority to oversee their respective zones. Certainly this made it more centralized and more effective tackling issues that affect urban centre

3.1.2 Establishment and Expansion

In **1913,** the post of president, honorary, was discussed which became an elective post in 1920. In **1926** the Cantonment board hired a Municipal Commissioner as the administrative head. The two Municipal boards consolidated in 1949 to form the Corporation of the City of Bangalore under The Bangalore City Corporation Act. The recently formed organization would oversee the tasks and enhance the government of cities.

3.1.3 Further Expansion 1980s-1990s

- **1989**: The Bangalore City Corporation was composed of only 75 wards but later it was extended to 87 wards...
- **1995:** The number of wards was raised to 100, and the corporation was renamed Bangalore Mahanagara Palike or BMP. These changes were accomplished by the physical planning and management of the city and its services to serve a larger municipal district.

3.1.4 Formation of BBMP 2007

The Karnataka Government passed a notification in the year 2007, proposing areas that fall under the existing Bangalore Mahanagara Palike (BMP), eight other urban local bodies and one hundred and eleven villages which come under the periphery of Bangalore city. This affected the decentralization process that resulted in the establishment of the Bruhat Bengaluru Mahanagara Palike (BBMP).

3.1.5 The BBMP Act 2021

In **2021**, the BBMP was formed under an updated legal system of what is known as the BBMP Act. This act targeted to have a better check-and-balance mechanism and reorganise the administrative machinery in the city. Overall now BBMP have 243 wards, 8 zones and in east zone there are 44 divisions and Domlur ward no 112 comes under Shanti Nagar division.

Key Developments and Innovations

- **Infrastructure and Urban Planning:** In its planning, BBMP has embarked on many projects to enhance infrastructure in the urban centres such as roads construction, sewage system and health facility among others.
- **Waste Management:** Among the important sub-sectors of environmental management SWM has been identified as one of the focal areas. To improve waste Segregation, segregation and processing, BBMP has initiated the construction of Dry Waste Segregation Centres (DWCCs) besides carrying out waste-to-energy projects.
- **Technological Integration:** The use of GPS integrated waste Segregation vehicles and mobile apps in tracking municipal services have increased efficiency and also

integrated mobile apple for complain people about SWM (app name: Namma Bengaluru) and accountability in urban management.

3.2 Organization Structure

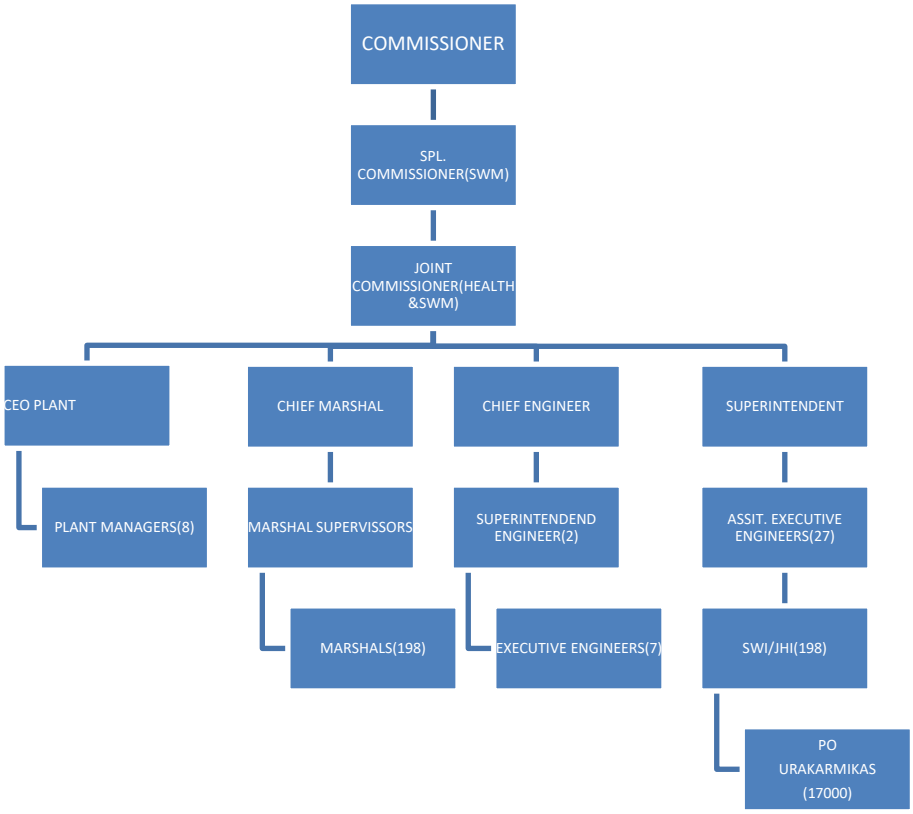

Figure 2 Organisation structure

East zone officers

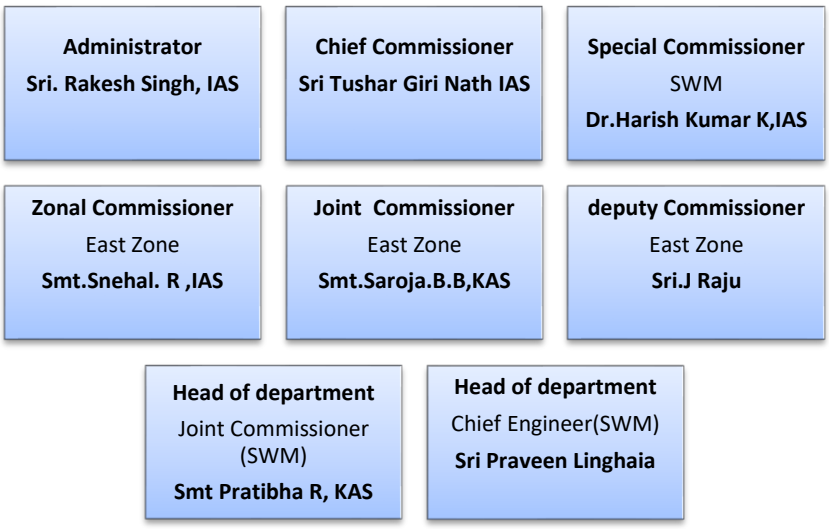

Figure 3 East Zone officers hierarchy

The image shows how a solid waste management commission is organized. The most important person, the Commissioner, is at the top, in charge of everything.

Below the Commissioner are a few key leaders, like someone specifically focused on waste management, another handling both waste and health, and a leader for the facilities that process the waste.

These leaders have different departments working under them. Some departments manage the day-to-day operations of the processing plants, while others oversee the infrastructure used to collect and process waste. There are also the workers who directly collect the waste

Dr. Harish Kumar, IAS, serves as the Special Commissioner (Solid Waste Management). His responsibilities include the preparation of policies in line with the "2016 Solid Waste Management Regulation" and SWM byelaws, additionally their implementation. He is also tasked with preparing policies for the welfare of Pourakarmikas (PKs), reviewing all zones regarding solid waste management, and introducing reforms and creating infrastructure for SWM in BBMP.

Based on the 2016 SWM Rules and Byelaws, the Zonal Commissioner makes sure that waste management policies are followed. They also take care of the well-being of Pourakarmikas (sanitation workers). The Zonal Commissioner checks that each division in their zone manages waste properly and builds the needed infrastructure.

All waste management initiatives in the city are supervised by the Joint Commissioner for Solid Waste Management in the BBMP. They make sure everything is running smoothly in each zone, check the work of different waste management teams, and review how well each area is sorting, collecting, and disposing of waste. They also take care of the well-being of sanitation workers and ensure that plans for their welfare are put into action by the officers.

The Superintendent Engineer (Head Office) for Solid Waste Management is basically the city's top trash guy. They make sure big plans for cleaning up trash happen, keep an eye on trash collection in neighbourhoods, and put new trash handling ideas into action.

The Executive Engineer (Head Office) for Solid Waste is a key player in keeping a city clean. They oversee big projects funded by the government, like building new ways to collect trash, better places to dump it, and even turning trash into something useful like electricity! They handle everything from recycling centres to making sure waste is disposed of safely, and even getting new machines to clean the streets.

The Divisional Assistant Executive Engineer is a key player in keeping their division clean. They make sure the head office's waste management plans are followed on the ground. This includes things like informing residents about new rules, making sure waste collectors follow the plan, and stopping illegal dumping.

Additionally, they engage the residents in their job. This could entail getting together for clean-up efforts, holding meetings with neighbourhood organizations, and launching educational initiatives.

There's some uncertainty about "PK welfare" in their role. It might involve managing waste in public parks or government buildings, keeping those spaces clean.

One important task is cleaning up areas with a lot of litter or illegal dumping. They'd work with sanitation crews and figure out why the dumping happens in the first place, maybe by adding more bins or signs.

Finally, they make sure waste gets sorted properly. This involves giving residents different bins for the respective types of trash and educating them on how to sort it. They also ensure the sorted waste goes to the right recycling or composting facilities.

The Ward Junior Health Inspector is basically the trash cop for their area. They watch out for people littering, dispose of garbage on wrong days, or not sorting their trash properly. Additionally, they see to it that everyone gets their garbage collected and assist residents in understanding the city's trash regulations. If they catch someone breaking the rules, they can even issue fines

3.3 Purpose and Mission

BBMP SWM department undertook a mission of implementing proper waste management measures followed by providing a clean environment in all Bengaluru.

3.3.1 Objectives of the mission:

1. **Cleanliness**: Ensure the achievement of a clean and healthy environment in the East Zone by properly addressing the Segregation, transportation, and disposal of wastes...
2. **Public Health**: The crucial element of effective waste management is the protection of public health since inadequate waste disposal may result in the spread of diseases and other health risks. To reduce such risks, the department has standardized the ways of handling wastes and observing sanitation procedures.
3. **Environmental Sustainability:** The department seeks efficient means to manage wastes within the city. This includes increasing the uptake of increased recycling,

composting and any other reduced methods of waste disposal than the conventional methods of either landfilling or burning.

4. **Community Engagement:** Attention shall be paid to communicate with residents, businesses, as well as other stakeholders for the success of the further waste management. HIV/AIDS prevention and control initiatives, for instance, involve the engagement of community members, creating awareness and partnering with any relevant organizations

3.4 Types of waste

Figure 4 Types of waste

3.4.1 Wet Waste

Bio-degradable waste, also called wet waste, is organic stuff that tiny organisms can break down into simpler, stable bits. This includes kitchen scraps like veggie peels and fruit leftovers, spoiled produce, meat scraps, garden waste like leaves and grass, coconut shells, and wood and leaf ashes.

3.4.2 Dry waste

Things that don't decompose naturally are referred to as dry waste and aren't inert street sweepings. This includes recyclable and non-recyclable materials, as well as combustible waste. Examples of non-biodegradable or dry waste include plastic items, aluminium cans, goods composed of Rexene and rubber, packaging materials, fabrics and clothes, and Styrofoam and thermocol.

3.4.3 Domestic Hazardous Waste

Domestic hazardous waste consists of household items that can pose risks like explosion, health hazards, or environmental damage. They may catch fire, react dangerously, or contain corrosive or toxic substances. Examples include aerosol cans, batteries (including car batteries), oil filters, car care products, bleaches, household cleaners, oils, solvents, cosmetics, insecticides, expired medicines, pesticides, herbicides, paints, glues, photographic chemicals, foam packaging, and products containing mercury.

3.4.4 Sanitary Waste

Sanitary waste refers to hygiene products like diapers, sanitary napkins, tampons, and bandages contaminated with blood, urine, and faces, as well as sharp objects like syringes and needles.

3.4.5 Construction & Demolition Waste

Construction and demolition waste refers to the waste produced during the building, renovation, or demolition of structures like buildings or civil constructions. This waste includes materials like concrete waste, debris, rubble, and any other waste generated during construction or renovation activities.

3.4.6 Mixed Waste

Mixed waste refers to the waste collected from designated public spaces such as parks, markets, roads, streets, gardens, and similar areas.

3.5 Daily Task Overview:

3.5.1 Waste Collection

Door- to-Door Collection

1. Route Planning: Waste collection routes must be designed so as to reach every household., slums, shops and other establishments in Domlur East Zone.

2. Auto Tipper Deployment: Wastes are collected from designated places using auto tippers. The allocation of auto tippers is determined according to the population of the locality and the rate of waste generation.

3. Daily Collection Schedule: Auto tippers go round the localities every day at certain times of the day to fetch the wastes. The collection schedule is publicized to the residents so as to enable them to dispose of their wastes at the right time.

4. Household Waste Collection: Auto tippers in this context do pick up waste from every household. Waste collectors help the residents in separating wastes into wet and dry categories.

5. Slum and Shop Collection: Besides households, wastes are also picked from the slum areas, shops, and any other establishment found along the collection route.

Bulk waste generation Collection:

1. **BBMP Records**: Get copies of documents and records of BBMP that pertain to the methods and frequency by which the Domlur East Zone gathers large trash or waste. This may comprise officially designed reports, guidelines, and circulars...
2. **Local Authorities and Officers:** Enquire with the local authorities and the local personalities like the zone commissioner, the ward officers and sanitation department officers to get detailed information from the management and practices used in handling bulk waste.
3. **Field Observations**: Go to the several fields to have real life contact with the processes of bulk waste collection. I notice the travel path of the waste collection trucks, the equipment used in collecting and transporting large volumes of waste, and any issues arising from waste collection.
4. **Interviews with Stakeholders:** Interviews with the commercial customers, marketers, hoteliers, and any other party engaged in the generation and disposal of bulk waste. Find out what they think about the effectiveness of the current collection system and what they would change about it.

Domestic hazardous waste Collection

- Trained contractors go to construction sites all around Domlur to collect construction and demolition wastes (C&D wastes). At the source, all waste materials such as concrete, bricks, tiles, wood, metal and any other scrap as well as debris are separated to ensure easy disposal.
- People assigned with the collection of wastes are properly trained, wearing their protective clothing and using the proper containers for these wastes, and collect wastes from registered houses and establishments at specific dates.
- To enhance the efficient disposal of waste, urge residents to sort waste generated at home by types biodegradable waste, non-biodegradable waste, recyclable waste and hazardous waste.

Market waste Collection

1. **Local Authorities and Records:** Approach the Bruhat Bengaluru Mahanagara Palike (BBMP) and the Solid Waste Management (SWM) Department of this city government body in order to obtain official records, reports or any other relevant documents pertaining to the aspects of waste collection in the market.
2. **Market Associations and Managers:** They should be contacted for a collection of first-hand knowledge of the volume of waste generated and the methods used in its collection in a typical commercial market. They may include information related to the amount and kind of rubbish that is produced, the timetable for its collection.
3. **Field Visits and Observations:** Pay socializing visits to Domlur East Zone's market sectors to inspect the Waste collection process. About the behaviours of the collection vehicles and employees, as well as the degree of segregation or disposal undertaken at each of the market locations.
4. **Interviews and Surveys:** Administer questionnaires or engage with informal discussion with the marketers or retailers in various markets, as well as responsible waste collectors in the markets. Gather existing practice and the experiences encountered; obtain ideas and impressions for process improvement.
5. **Community Feedback Sessions:** Attend meetings hosted by market authorities or governance agencies. These forums expose an ability to solve the issues of waste management commissioned in specific market areas and collect data from the interested parties.

6. **Media Reports and Publications**: Read through local newspapers, magazines, journals and any other literature material that discusses waste management efforts and resurgence in the market areas. These sources may focus on a certain subject, best practices, or social projects regarding market waste collection.
7. **Online Platforms and Websites:** he online sources that may include the official website of BBMP or different executive market associations or other community sites exist. These apps may be informative on any developments in waste management services, changes in the collection schedules, and other important contacts

Sanitary waste Collection

This goes hand in hand with the involvement of experts who are always working in the field, collecting sanitary waste from the residences, flats, and other housing units within Domlur. Residents will not be much of a bother as they dispose of their waste materials.

Sweeping and Collecting Park Leaves and Road Debris

1. Sweeping Schedule:

- In the proposed study, BBMP SWM East Zone Domlur dispose SWM in a systematic manner by following a predetermined sweeping timetable for parks and roads.
- The sweeping frequency may also differ depending on the degree of area usage, i. e. the number of people accessing the particular area or floor, seasonal variations, or festivities and celebrations among others.

2. Park Cleaning:

- SWM East Zone Department the trained sanitation workers are assigned to only those parks and green area which falls under the jurisdiction limit of BBMP of Domlur.
- Park cleaning entails cleaning pavements and open areas where individuals can unwind and exercise, walking with broom to clear the floor of fallen leaves and any debris.
- Such concerns are taken with regard to playgrounds, benches, and other places where people relax or spend their time also with the overall landscaping.

3. Leaf Collection:

- Specialists who use rakes, sweepers, and mower to clear fallen leaves are employed to clear them from park area.

- Leaves and other such biodegradable and recyclable wastes are collected and put into heaps or disposed of in bins for further use depending on the arrangements made along with implementation of BBMP.

3.5.2 Segregation

1. Door- to – door segregation

- Some of the measures that have been put in place by BBMP include the undertaking of awareness creating trips to ensure that the residents adhere to the 3R's and source segregation.

- People become aware of three main categories of waste products, namely hazardous waste, dry waste, wet waste and the need to sort the waste products.

- Segregation bins are also delivered to households, often with a colour coding which is easier to distinguish (e. g. These included red for recyclable items, black for hazardous waste, brown for non-recyclable waste, pink for glass, yellow for food waste, grey for general waste, orange for confidential waste, green for wet waste, and blue for dry waste.

Segregation at Source:

- All people sort their waste which includes the wet waste and dry waste at source in colour coded bins.
- Y Wet waste problems are deteriorating because they contain biodegradable organic materials like leftover food materials and garden stuff.

Pourakarmikas' Role:

- Pourakarmikas (sanitation workers) are directly involved and are capable of ensuring proper mechanism of segregation at source.
- They sensitize people and compel them to follow the right channel in classifying waste during collection trips.

2. *Bulk waste generation Segregation*

BBMP categorizes specific properties like hotels, restaurants, markets, offices, and marriage halls etc. as BWGs since they generate large amounts of waste.

Registration and Notification

- BWGs must follow the registration procedure with BBMP and inform the organization that they operate as bulk waste generators.
- At the point of registration, the BWGs are provided with policies and procedures on how to separate and dispose wastes.
- Segregation Infrastructure Setup:
- BWGs put in place segregation structures to aid in waste separation on their grounds.
- Personal waste containers are allocated for every category of waste, wet waste, the dry waste and hazardous waste.

Training and Awareness:

- The various BWGs are educated through capacity training and sensitization sessions facilitated by BBMP.
- It encompasses the methods of properly disposing wastes, including segregation, management and disposal methods and the regulations guiding waste management.

On-Site Segregation:

- During waste generation, individuals such as staff and employees are supposed to sort their waste, putting it in the relevant container depending on the category it belongs to.
- Organic waste such as wet waste is separated from non-organic or dry waste which include; plastics, cans, papers, cups and strings.
- Namely, BBMP routinely monitors and audits facilities at BWG with the aim to ascertain compliance to the waste segregation standards.

- These consist of the examination of physical barriers that help to maintain segregation and the identification of proper disposal of wastes.

Reporting and Record-Keeping:

- Hence, BWGs preserve written records on waste generation and segregation activities as per provisions of the BBMP.
- Annually or quarterly reports on waste segregation and disposal are separately prepared and given to the BBMP authorities for assessing them.
- Consequently, if waste segregation regulations are not followed, BBMP will impose penalties or fines for any non-compliance.
- BWGs who do not manage to separate wastes or dispose them appropriately are liable to be penalized.

3. (C&D) waste segregation

Identification and Source Separation

Construction and Demolition waste is the one generated as a result of construction, major renovation or demolition of large infrastructures. This material comprises concrete, bricks, wood, metal, plastics and other debris.

All the builders, contractors and site supervisors generate C & D waste should segregate at source i.e., construction sites.

On Site Segregation facilities:

As per the BBMP notification, separate bins or demarcated areas for different types of C&D waste should be provisioned at construction sites.

Use of different colour containers or bins to separate out recyclable and non-recyclable wastes.

Identifying the Recyclable Material

The recyclable materials like concrete, bricks, metal and wood can be identified in the rubble itself which is then separated and processed acceptably for recycling. Recyclables are sorted and kept separate from non-recyclable waste.

Domestic hazardous waste Segregation

Identification and Collection

- Domestic hazardous waste refers to the household products that have danger of bursting into flames, being detrimental to healthy, or harming the environment. These may consist of aerosol cans, batteries, oil filters, hospital and commercial grade cleaning fluids, paints and pesticides.
- Offensive collection services provide and pick domestic hazardous waste, volunteered with other general waste from homes in an effort to avoid interference and correct management.

Segregation at Source

- It is households that are required by law to sort out domestic hazardous wastes from other types of domestic wastes at source.

- Special containers might be used to gather the contaminating waste to ensure it is separated in its collection.

Secure Packaging

- Dissolvable items should also be carefully packed so as to ensure that they won't mix with other wastes, leak or perhaps break.

- A few products like batteries and chemicals may be packed in a sealed bag or covered with protective layer so that the passengers are shielded from chemical corrosive substances.

Labelling and Identification

- It's essential to make hazardous waste containers or bags recognizable, or to use some labels or marks which would inform that hazardous materials are stored inside.
- They may contain warning phrases or symbols that should inform the handlers of waste regarding any dangers they can experience when interacting with waste.

Safe Handling and Transport

- The handlers of hazardous wastes are personnel trained to handle and transport the wastes wearing appropriate PPEs.

- Flammable waste should be handled carefully in order to avoid adverse unintended exposure to dangerous chemicals or substance particularly when collecting and transporting the wastes.

Segregation at Collection Points

- Household hazardous waste is temporarily stored at various collection bins or transfer stations to enable sorting.

- Cherry pickers employed by waste management organisations, separate hazardous waste materials from other waste materials to prepare them for adequate disposal and further treatment.

Non-Recyclable Waste Separation

- Materials referred to here as 'other wastes,' including plastics, packaging, and mixed debris, are also collected and sorted out from the recyclable ones.

- These include chemicals like asbestos, lead-based paints, or chemical residual whose management is done bearing in mind the laid down safety procedures.

Hazardous Material Handling

- A list of materials deemed dangerous to handle, store or dispose is put down and the appropriate tags put on containers housing such specimens, and their handling, storage and disposal follows the provisions of laws, guidelines and norms as provided by the law.
- Regarding guidance and assistance for builders and contractors the BBMP may give direction and recommend the proper methods for disposing of hazardous C&D waste.

4. Market waste Segregation

Identification of Market Waste

Market waste comprises of the vegetable and fruit peels, cobs, and other waste materials found within the market or within the surrounding streets where mainly the eatery businesses are located.

Segregation Infrastructure Setup:

- He/she said BBMP officials have placed separate bins or containers at some chosen places in the market to ensure segregation of different types of waste.

- Waste banks that are normally employed have a distinction between the ones for wet (organic) waste and the dry waste ones.

Vendor Education and Awareness

- To guarantee the application of the segregation principles, BBMP periodically organizes awareness raising and training events among the buyers and sellers of the markets.

- Shall provide instructions to the vendors and encourage them to store their wastes separately to prevent littering the surrounding environment.

On-Site Segregation:

- At the market stalls, waste is sorted by the vendors into the respective bins which are that of wet waste and dry waste bins.
- Bio solids such as leftovers, peelings, inconsumable, and plant materials are placed in wet waste bins meant to contain organic wastes.
- Non-organic waste including plastic paper, cardboards and anything that is non-biodegradable is put into bins for dry waste disposal.

5. Sanitary waste Segregation

Identification and Collection:

- The sanitary waste includes diapers, sanitary towels, tampons, cotton/bandages and other biomedical waste that are soaked in blood, urine or faces.
- Similar to other types of garbage, sanitary waste is separated from other waste sin order to avoid mixture and to facilitate proper management.

Segregation at Source

- Households, institutions, and, healthcare facilities are to isolate sanitary waste from other waste at source at the point of generation.
- Proper arrangements of sanitary waste collection may involve the use of several bins or containers that allow for sorting.

Secure Packaging:

- Products such as diapers, sanitary towels, napkins, and bandages should be disposed in enclosed containers, sealed, unbroken or tied to minimize chances of them melting and causing a mess.

- Sanitary waste bags or biohazard bags or pouches may be used to minimize the possibilities of contacting fluids & wastes that have pathological or infectious contents.

Labelling and Identification

- Sanitary waste containers or bags should be yellow and therefore, different from general waste container-colour codes for wastes.

- The tags that suggest that a particular container is contain biomedical waste or other hazardous items can be attached to the containers.

Safety in Storage and Transportation

- Sanitary waste is carried by trained workers wearing proper protection as provided for in the EPZ regulations on Occupational Health and Safety.

- Sanitary waste should therefore be handled in a proper manner to avoid direct contact with the pathogens or hazardous substances which is likely to occur during pick up and disposal.

- This segregation is done whereby some personnel separate itself from the other in the collection points.

- Household, institutional or health facility sanitary waste is transported to properly designated collection centres or transfer stations where it is sorted.

- The starting point for supervision is where waste management personnel are capable of separate sanitary waste from other waste forms to allow for proper disposal and treatment.

Sweeping and Collecting Park Leaves and Road Debris

Leaf Sweeping Operation

- Sweeping is a function that includes the clearance of leaves and debris on parks and roads by mechanically enhanced sweepers, brooms, or air blowers.

- Sweepers and workers regularly sweep the scheduled specific area such as corridors, streets, and lawns to gather piled up leaves and organic matter.

Collection of Leaves

- After leaves have fallen to the ground, they are gathered with common, rakes, shovels or mechanical loaders.

- Leaves ought to be properly packed on the ground where they fall so that workers can easily pick them and put in more manageable piles or heaps.

Segregation at Collection Points

- For instance, at collection centres, the collected leaves are sorted together with the material components and methods of disposal.

- Leaves are categorized into sub groups such as green waste meaning they are an organic waste, mixed waste meaning that they have other waste materials, and hazardous waste in-case they have other pollutants or chemical material that poses a threat to human life.

Segregation Criteria

- Green Waste: Goods derived from the plants that are biodegradable and can thoughtfully be recycled in composting or as mulch materials are categorized as green waste. This category includes samples of plants that have no contact with any other substances other than water during the extraction process.

- Mixed Waste: When the green waste is mixed along with additional waste materials like glasses, papers, plastics or any other non-biodegradable material then it is categorized as mixed waste. The schedule should factor these materials in that they need to be dealt with in a different manner from other waste products.

- Hazardous Waste: Any waste material, solid waste, liquid waste, excessive gases, contaminated leaves or any material that contains hazardous substances or pollutants or chemicals can be termed as hazardous waste. Such materials require additional care when being transported and dealing with; hence the proper procedures are observed when it comes to disposal.

3.5.3 Transportation process overview:

Door- to-Door Collection transportation

Loading of Waste

- We note that waste collection crews move from one house to another within the existing routes with an intention of picking the waste.

- The urban waste is collected and disposed of by using collection vehicles and since it is segregated some of the wastes are physically loaded by the crew members into the compartments of the collection vehicles.

Compaction and Storage

- With compactors though, the waste collected may be compacted within the vehicle to get as many compaction possible within the vehicle to delay unloading.

- Other storage units which apply to waste storage compartments are well closed in order to avoid any spilling or leaking during the transit process and consequently reduce adverse impacts on the human health and environment.

Transportation to Transfer Stations or Processing Facilities

- The waste collection vehicles, once fully stocked, then carry the collected waste to either transfer stations or processing facilities.

- Transfer stations are allows of waste from collections routes and which transfer the wastes to the final disposal or processes points.

Unloading and Disposal

- At the transfer station or processing facility, the vehicles that collected the waste then tipped it in a designated area for further processing, sorting or disposal.

- According to the type of waste and specific waste disposal policies used by different regions, collected waste is often taken to composting facilities, recycling plants or landfills, or waste to energy plants for their final disposal or for recovery.

Return to Collection Routes

- Following disposal of this waste, the district collection vehicles may go back to their collection routes to embark on other consecutive rounds of waste collection from door to door.

Bulk waste generation transportation

Waste Collection Vehicles:

- Self-contained compactors or trucks with large receptacles for holding the volumes of waste are used to collect the waste from the source generators in special trips.
- These automobiles have compartments or bins for general storage of various types of wastes such as wet wastes, dry wastes, and hazardous wastes.

Scheduled Collection Routes:

- Collection routes are pre-determined according to the geographical area of dense concentration of large waste producing institutions, the rates of waste production among these institutions and any other realistic considerations that could arise.

- Collection schedule also guarantees regular collection of waste from the bulk generators to avoid waste build up and eventual.
- From this, it is clear that collection schedules help in maintaining the right time and frequency through which waste is collected from the generators to avoid waste.

Loading of Waste:

- Municipal workers or waste handlers transport the separated bulk waste to the collection vehicles by using their hands to pick and place it in the proper compartments.

- Finding ways of handling and ensuring that wastes are well disposed so that they do not get mixed or cause harm of any sort to the surroundings during transportation is well observed.

Transportation to Processing Facilities

- After being filled to optimum capacities of the contained waste, the waste collection vehicles then convey the bulk waste to processing centres or transfer stations.

- Each composting facility or waste treatment plant may be different; however, such plants may be composting plants, recycling plants, or plants dedicated to the disposal of hazardous wastes, based on the kind of waste that was received.

Unloading and Disposal:

- These disposal centres or stations are also known as processing facilities or transfer stations, where the collected waste in bulk is removed from transportation vehicles for sorting, processing, or disposal.

- The actions can include recycling, composting, or disposing hazardous waste in line with the laid down national and regional measures.

Return to Collection Routes:

- Sweeping and washing trucks may also go back to the route to pick more waste in subsequent trips for those garbage producers who generate large quantities of waste.

(C&D) waste Collection transportation

Scheduled Collection Routes

- Consequently, based on the places of the construction and demolition sites, the quantity of waste produced and other factors that influence the collection routes, BBMP maps out its collection programs and schedules accordingly.

- Collection routes are established to provide a systematic and scheduled method of picking C&D waste to avoid a build-up and the creation of a mess in the environment.

Collection Mechanism

- Independent service providers or those personnel employed by BBMP are to take the lead in collecting C&D waste according to a timetable and specific routes assigned.

- Collection crews employ specialized vozdukhans, dump trucks, roll-off containers or skip bins to collect the waste material mostly from construction and demolition presents.

Loading of Waste

- The wastes after segregation are collected individually by the waste collection crews and placed directly into the appropriate sections or bins of collection vehicles.

- There is also observance made in order to guarantee proper storage and identification of different types of waste to avoid spilling and for transportation purpose.

Transportation to Processing Facilities

- Collection vehicles then take the C&D waste to disposal sites, recycling stations, or landfills depending on the status of the disposal area.

- Treatment facilities could be C&D waste recycling plants, debris recycling centres or disposed Sanitary landfills based on the specific waste material that is being processed.

Unloading and Disposal

In the plants or at the landfills, C&D waste is offloaded from the transport vehicles for more processing, sorting or disposal.

- There are not many waste management measures can include proper disposal of non-recyclable materials in specialized sites or areas, otherwise disposal can entail reusing of such material as concrete and metals, crushing of debris for use in construction as aggregates among others.

Return to Collection Routes

- Depending on the number of rounds done at a construction or demolition site, collection vehicles can then move to their respective routes to collect wastes during the next round.

Domestic hazardous waste Collection transportation

Loading onto Transport Vehicles

- Often, the hazardous waste is packed carefully into the transport vehicles, so that incompatible types like flammable and explosive hazardous wastes are not grouped together.

- They have safety arrangements such as containment system to cover spillage or leakage on the vehicles during transportation.

Transportation to Treatment Facilities

- Hazardous waste is carried to duly licensed treatment, storage, and disposal centres (TSDs).

- In transit the vehicles use routes and adhere to laws concerning handling of specific goods especially the hazardous ones.

Unloading and Treatment

- The receiving of hazardous wastes takes place at the TSDF under a specific manner that is followed by the providers.

- Hazardous waste can be treated by certain processes like incineration, chemical neutralization or stabilization based on the kinds.

Final Disposal

- After treatment, the remaining hazardous waste is buried in hazardous waste disposal sites or disposed of using other officially approved techniques.
- Techniques for disposing of hazardous materials eliminate the possibility of their effects on the environment and human health.

Market waste Collection transportation

Collection Mechanism

- The respective waste collection service providers or the personnel employed by BBMP are supposed to lift the market waste in accordance with the time schedules and health routes planned and assigned for each respective market.

- Collection crews employ compactor or storage trucks to empty markets that have collected waste in bins or bags.

Loading of Waste

- The staffs of waste collection also ensure that solely market waste is collected and placed in appropriate compartments of the waste collection cars.

- Proper handling of waste is accomplished while considering the disposal of different wastes in a manner that will reduce cross contamination and easy transportation of wastes.

Transportation to Processing Facilities

- Collection vehicles then take the market waste and deposit it in processing centres or transfer stations.
- Integrated waste management facilities may include recycling centers, composting plants, or any other waste-handling facility, contingent on the kind of wastes gathered.

Unloading and Disposal

- Market waste is received and temporarily stored in holding or transfer sheds or stations at the various processing units or before transportation for treatment or removal.

- Managing waste depends on the category of waste involved, though steps are taken for reusing, recycling, or disposal of waste appropriately in accordance with the laid down guidelines.

Return to Collection Routes

- After dropping off the waste, the collection vehicles can go back to sweep the routes for other consecutive trips in the market zones.

Sanitary waste Collection transportation

Scheduled Pickup Routes

- The BBMP or other contracted agencies determine fixed routes and times for the collection of the sanitary waste from particular areas such as residential areas, hospitals, clinics and public toilets among others.

- Techniques are laid down to ensure its schedule for collection of sanitary waste frequently and prevent stock piling, hence the cleanliness of the community.

Loading of Waste

- The sanitary waste is collected in a collection car using either mechanical equipment or by hand manipulation.

- Precautions are regularly made to ensure that waste receptacles are closed and packed properly in the collection vehicle such that they do not tip over or mix with other wastes while in transit.

Transportation to Treatment Facilities

- The sanitary waste is then collected by vehicles which take the waste to facilities that are capable of handling biomedical waste.

- Hospitals and treatment facilities may consist of large biomedical waste treatment centres where the waste is treated through sterilization or some other form of disinfecting and treatment before the waste can be properly disposed.

Unloading and Disposal

- Sanitary waste is then offloaded for treatment from collection vehicles once it has arrived at the treatment facility to its specific processing zones.

- In order to protect human health and the environment, workers performing waste management tasks must follow certain criteria for managing and disposing of biological waste.

Treatment and Final Disposal

- Biomedical waste is treated by various methods based on the type of waste and the available technologies at the treatment facility like autoclaving, microwaving or incineration.

- Specific to waste leftovers that are produced as treated waste, sterilized or ash residues of wastes are disposed in compliance with the requirements of the law through landfilling or any other appropriate method of disposal.

Sweeping and Collecting Park Leaves and Road

Sweeping Schedule:

- When the process of the East Zone, BBMP SWM Domlur sweeps parks and roads regularly as per schedule to keep them clean and hygienic.

- The sweeping frequency may therefore vary based on factors such as foot traffic, seasons and other occasions within a given year.

Park Cleaning:

- Sanitation workers undergo training and are supervised to work in different parks and other green areas which come under BBMP SWM East Zone, Domlur.

- Park cleaning as one of the social responsibilities of civil society organization entails sweeping of pathways, open space, and recreational facilities to clear perhaps falling debris such as leaves, litter among others.

- Particularly important are the issues of lighting, recreational or playground areas, and any structures and plantings intended for visitors.

Leaf Collection:

- Special teams are usually set with the responsibility of cleaning up the park grounds using rakes, brooms and leaf blowers with a view of picking the fallen leaves diligently.

- Leaves collected in different department and areas are thrown in heaps or bins are collected and transported for disposal or for composting as per the existing norms of BBMP.

Road Cleaning:

- Street cleaning exercises are usually undertaken on roads and streets where there is aggregation of dust, dirt as well as other waste materials on the streets.

- Mechanical way or manual sweeping is conducted with an aim of sweeping roads, walkways, curbs, and medians.

- To create both beautiful and safe landscapes, special attention is paid to areas with high traffic volumes, crossings, and crossroads.

CHAPTER-04

4. Brief Data Analysis

4.1 Project description

For my social immersion project, I chose the Bruhat Bengaluru Mahanagara Palike (BBMP) Solid Waste Management (SWM) department of East Zone, specifically with the Domlur Ward. Improper disposal of the solid waste would cause serious impact on the health and environment thus it needs to be improvised and train the individual households and the bulk wastage generating market places. This project was to expose me to practical experiences on the operations & challenges of municipal solid waste management and to allow me to gain field & office exposure. Additionally to adopt the best foreign solid waste management strategies into our management. Improvement and guidance to the individuals in the wastage segregation from individual households

- Collection, segregation and transporting the wastage on daily basis.
- Volunteering in community improper disposal of solid waste.
- Reviewing data concerning generation, collection, and disposal of wastage.

This ensures that there is a constant check on how efficient the waste collection routes are.

Working with the RWAs and NGOs for better enforcement of rules and regulation in the neighbourhoods.

4.2 Data Analysis and Observations

4.2.1 Quantitative Data Analysis:

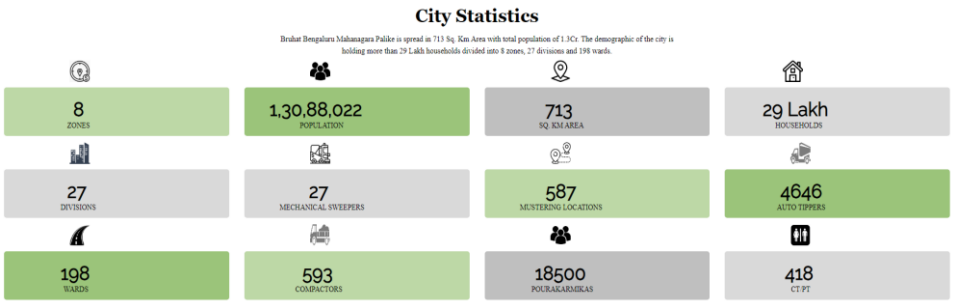

Figure 5 City Statistics

The Bruhat Bengaluru Mahanagara Palike, which is the governing municipal corporation of Bangalore city in India. Here is a breakdown of the information presented in the image: Here is a breakdown of the information presented in the image:

Zones: There are eight distinct districts for administrative convenience.

Population: It has a population of 130, 88,022 people or 13 point 08 million as per 2011 census.

Area: It occupies an area of 713 square kilometres in size, which signifies a large area being occupied by the city.

Households: As of the Indian census of 2001, there are 2. 9 million homes in the city.

Divisions: It comprises 27 divisions that work collaboratively in achieving the goals of the city.

Mechanical Sweepers: Mechanical cleaning uses 27 sweepers to perform the cleaning activity.

Mustering Locations: Processing and identification landmark: There are 587 points or places with the word 'mustering' (that is probably the place where those people, involved in work, gather or sign in).

Auto Tippers: It also shows that there are 4646 auto tippers, the type of vehicle used in garbage collection and transport.

Wards: The city has been split into some twenty-one hundred and ninety-eight wards.

Compactors: At the SAGG, there are 593 compactor; it is a vehicle or machine used in compaction of waste to reduce its volume.

Pourakarmikas: Implements available in a3 or a4 are 18500 Pourakarmikas (sanitation workers).

CT/PT: Community toilets and public toilets present amounts to 418, abbreviated as CT/PT.

These figures provide information regarding the parts and endowments of the actual supervision and sustenance of the city with emphasis on factors like population, area, division and sanitation.

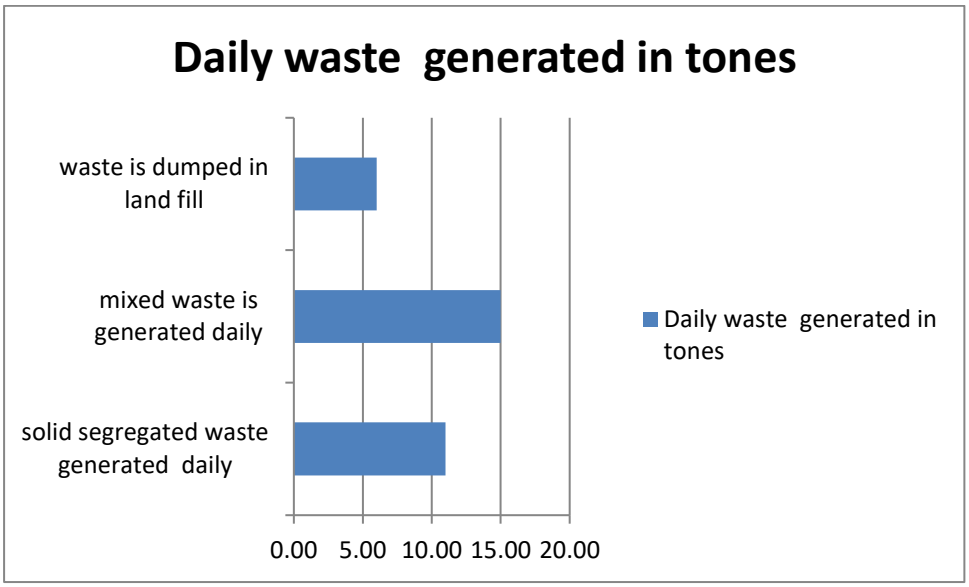

Figure 6 Daily waste generations

The bar chart named "Daily waste generated in tonnes" demonstrates the amount of waste produced every day in different divisions. This is what each category of the chart represents:

Waste is disposed in dump:

Approximately six tons of garbage is thrown into landfills per day.

Every day mixed waste is created:

About fifteen tons of mixed trash is produced on a daily basis. These include recycling materials and others which have not been separated yet.

Daily solid segregated wastes are generated:

A little over 8tons of solid segregated wastes are produced every single day. This refers to refuse that has been classified as recyclables, organic matter or any other group.

This graph summarises daily volumes and types of wastes, highlighting the fact that there's more mixed waste than segregated and landfill-bound ones.

Total population	**15000**
Total waste (in tonnes)	32.00
The per capita waste generation is approximately(in tonnes)	0.00213

Figure 7 Volume of waste in tonnes

From the above data we will see that the total population is 15,000, and the city generates 32 tonnes of waste daily.

Per capita waste generation =32tonnes/15000people =0.00213tonnes/person/day

This means each person generate approximately 0.00213 tonnes per day of waste.

In this graphs we will see this feels like data on waste management in some specific community or municipality.

- 9-12 tonnes solid segregates waste generated daily – This shows that between 9 and 12 tons of solid rubbish is sorted out every day. Segregated waste, typically, means it has been separated into various categories like recyclables, organic waste and non-recyclables.

- 15 tonnes mixed waste is generated daily – Approximately 15 tons of mixed garbage per day. Mixed wastes refer to those wastes that have not been sorted out and they comprise a mix of different types of materials.

- 6 tonnes waste is dumped in landfill - Around Approximately six tons of garbage are brought into a landfill each day. Such trash mostly consists items that cannot be recycled or composted hence need to go to a landfill.

- 19 Waste Collecting Vehicles - There are nineteen vehicles set aside for the purpose of collecting garbage within this particular region; these vehicles likely collect both segregated as well as mixed waste from various collection points in the surroundings.

- 1 Composter – The community has only one unit or facility for composting purposes. A composter is useful for changing organic rubbish such as food leftovers together with garden residues into compost which could be used as fertilizer.

4.3 On-going Consideration

- **Practical Knowledge:** Able to recognize the challenges faced in waste management, both organizational and practical, alongside community issues.

- **Data Analysis Skills:** Improved my analytic skills and capacity to make conclusions, specifically in regard to the efficient organization of waste picking routes and efficiency of segregation.

- **Communication Skills:** Developed an enhanced knowledge of working with different actors of the local community, including city administrators and other authorities, the inhabitants, and leaders of the community.

- **Problem-Solving:** As for the practical benefits, it allowed to develop the problem-solving skills by analysing the weaknesses of the operations and define the necessary measures.

The theoretical concepts and practical knowledge as well as problem-solving that I have gained within this project have prepared me well for the future within public administration and urban planning.

CHAPTER-05

5. DISCUSSION AND CONCLUSION

5.1.1 Social Project Experience

I have taken the BBMP Solid Waste Management (SWM) as my social immersion site in the East Zone – Domlur which was quite insightful for me. While engaging in this project, I have effectively availed a perfect understanding of the factors that are associated with waste disposal in urban areas as well as the importance of getting the community's contribution towards enhancing the environmental sustainability in the right perspective. Interacting with the community directly and working with officials of the BBMP and the NGOs involved in waste collection and segregation has broadened me a lot and helped me understand many practical aspects of how a municipality functions.

5.2 Learning's and Challenges

A major finding of this project is that waste management initiatives should be undertaken in a stepwise and iterative manner where the effectiveness of the measure is regularly assessed for any signs of improvement. Thus, apart from the knowledge of the specific industry, I have benefited from the acquisition of relevant skills such as project management, data analysis, and building relationships with the community. Realising the issues that many municipal organisations have to juggle and dealing with limited staffing and funding sources has been insightful and has served to show that fresh ideas and careful management of resources is paramount.

The problems that were faced comprised of the coordination of other people and organizations involved in the implementation of the intervention, dealing with the residents' reluctance to change their behaviour, etc. The development of techniques for resolving these issues have improved my problem-solving abilities. And difficulty in finding primary data such as challenges faced by BBMP. That has given me a broad perspective on patience, perceptiveness, and the value of enduring and effectual approaches within public service projects.

5.3 Future Learning and Observations

The topics in which I have a special interest to learn more are such as; project management strategies which are more specialized as well as go into more detail on how accounting works in the municipal setting. It is there that I have become interested in observing the policy implications of waste management practices since it also influences the regulation of resource management and managerial strategies for sustainable urban advancement. Also, some of the topics that I will want to explore further include the managerial implications of pricing and resource distribution as well as the considerations of delivering quality and distinctive services to clients.

- Successfully helped in managing efficient routes for garbage collection thus increasing efficiency with regards fuel consumption.
- Conducted several community sensitization meetings which were greatly appreciated hence contributing to a higher level of waste separation.
- Successfully developed effective working relations with the BBMP staff and the community members and counterpart.
- Faced difficulties due to differences in data collection methodologies that may time and again present analysis barriers.
- There were locals who vehemently objected to the modifications being imposed to their waste management practices..
- At times there was restricted access to materials and equipment which hindered the execution of pending changes.

5.4 Suggestions

- Increased Community Engagement: Daily and weekly cleaning activities to ensure that waste is properly sorted and discarded in the appropriate disposal bins.

- Technological Integration: Expanding tracking systems of waste collection to enhance responsibility and productivity.

Use of technology for use of waste, which was used by other counties such as:

"Technology to effectively extract fuel and electricity from biomass waste"

- Paper and other combustible trash with low moisture content are among the waste materials produced in cities and towns.

- Debris such as plastic and wood, as well as waste materials that contain a lot of moisture, like sewage sludge, manures, biomass, and waste from food production.

"Hita City, Oita (Use of biomass with high moisture content)"

Farms gather sludge from an effluent treatment facility, kitchen waste from residences and businesses, and hog manure, and treat them all together at a methane fermentation facility.

.

"Kyoto City (production of fuel oil (BDF) from oil waste)"

Cooking oil waste recovery system is structured to manufacture BDF, which is used as fuel for city-operated buses and garbage collection trucks.

"Technology to produce high-grade recycled PET resin and recycle products"

- Japan has started collecting PET bottles, food trays, and cans separately for recycling purposes in the production of new items, in accordance with the 3R policy (Reduce, Reuse, Recycle).

- The law aimed at promoting sorted collection and recycling of packaging and containers. This regulation creates a system where waste paper, glass, and plastic containers as well as packaging

"Medical waste disposal technology"

- A portion of the trash produced by medical facilities could be infectious. Needles used on hepatitis patients ultimately resulted in the infection of medical workers with hepatitis B.

- To ensure worker safety and prevent cross-contamination, medical waste is disposed of using plastic containers, cardboard boxes, and metal containers.

- Some of the furnaces that are utilized as medical waste incinerators to control the emission of dioxin are vertical furnaces that burn waste entirely and safely, kilns furnaces, and gasification furnaces.

"Municipal waste incineration technology"

- The Tokyo-based Shibuya Incineration Plant has successfully met stringent gas emission standards for NOx, SOx, smoke, dioxin, and other pollutants.

- Policy Advocacy: As a result, enhancing rules and providing rewards for environmentally friendly waste management techniques.

5.4.1 Conclusions:

In conclusion, my social project with BBMP SWM has been an enriching learning opportunity for me where I have gained hands-on experience and perspective towards solid waste management in urban cities. The chance to work on a project with a focus on environment and community has been rewarding and I would like to thank my supervisors and fellow co-workers for their help. I do believe that I am ready to use my new experience and knowledge in the future.

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- https://bbmp.gov.in/home

9. WEEKLY REPORT

WEEK 01

Name of the Student and Roll Number	PRAFUL P18UZ22M015036
Name of the Topic	SOLID WASTE MANAGEMENT IN BENGALURU WITH REFERENCE TO BENGALURU EAST ZONE
Period of the Report (Week 1st / 2nd / 3rd / 4th)	1st week
Activities are undertaken during the week (Students should briefly narrate the activities as bullet points)	a. Title was approved b. Objectives was framed c. Abstract was prepared
Details of field trips undertaken (if any) and summary of results of such trips	Nil
Learning Points acquired from the above activities	Selecting of titles and how to frame objectives and abstract
Plan for the next week Any other point	Starting with the chapter 1 Introduction

WEEK-2

Name of the Student and Roll Number	PRAFUL P18UZ22M015036
Name of the Topic	SOLID WASTE MANAGEMENT IN BENGALURU WITH REFERENCE TO BENGALURU EAST ZONE
Period of the Report (Week 1st / 2nd / 3rd / 4th)	2nd week
Activities are undertaken during the week (Students should briefly narrate the activities as bullet points)	1st chapter is completed and started with 2nd chapter
Details of field trips undertaken (if any) and summary of results of such trips	Nil
Learning Points acquired from the above activities	How to frame social relevance of the topic
Plan for the next week Any other point	Should complete 2nd chapter and start with 3rd chapter.

WEEK-3

Name of the Student and Roll Number	PRAFUL P18UZ22M015036
Name of the Topic	SOLID WASTE MANAGEMENT IN BENGALURU WITH REFERENCE TO BENGALURU EAST ZONE
Period of the Report (Week 1st / 2nd / 3rd / 4th)	3rd week
Activities are undertaken during the week (Students should briefly narrate the activities as bullet points)	Done with 3rd chapter and stating of data collection
Details of field trips undertaken (if any) and summary of results of such trips	Visit to dumping ward to collect primary data.
Learning Points acquired from the above activities	Learned how to interact with peoples and there I meet a inspector SWM(BBMP) and President of RWA and gained knowledge about challenges and field work
Plan for the next week Any other point	Analysis of primary data

WEEK-04

Name of the Student and Roll Number	PRAFUL P18UZ22M015036
Name of the Topic	SOLID WASTE MANAGEMENT IN BENGALURU WITH REFERENCE TO BENGALURU EAST ZONE
Period of the Report (Week 1st / 2nd / 3rd / 4th)	4th week
Activities are undertaken during the week (Students should briefly narrate the activities as bullet points)	Done with the analysis and completed the chapters
Details of field trips undertaken (if any) and summary of results of such trips	Visited dumping ward and meet Marshals and Pourakarmikas
Learning Points acquired from the above activities	Learned how SWM department operates Learned how to prepare a project and to analyze the data
Plan for the next week Any other point	Checking of plagiarism and should submit the report

DrillBit

DrillBit Similarity Report

6	36	A
SIMILARITY %	MATCHED SOURCES	GRADE

A-Satisfactory (0-10%)
B-Upgrade (11-40%)
C-Poor (41-60%)
D-Unacceptable (61-100%)

LOCATION	MATCHED DOMAIN	%	SOURCE TYPE
1	ir.canterbury.ac.nz	1	Publication
2	www.slideshare.net	<1	Internet Data
3	core.ac.uk	<1	Publication
4	moam.info	<1	Internet Data
5	pdffox.com	<1	Internet Data
6	www.nrc.gov	<1	Publication
7	docplayer.net	<1	Internet Data
8	susproc.jrc.ec.europa.eu	<1	Publication
9	cag.gov.in	<1	Publication
10	udsspace.uds.edu.gh	<1	Publication
11	Sustainable acoustic and thermal insulation materials from elastomeric by H-2011	<1	Publication
12	Constraints to the implementation of IPM programs in the USA A course outlin by Kimberl-1986	<1	Publication
13	mohua.gov.in	<1	Publication

14	online-journal.unja.ac.id	<1	Publication
15	www.e-yearbook.com	<1	Internet Data
16	www.uniindia.com	<1	Internet Data
17	academicjournals.org	<1	Publication
18	Thesis Submitted to Shodhganga Repository	<1	Publication
19	A healthcare logistic network considering stochastic emission of contamination by Nikzamir-2020	<1	Publication
20	sportdocbox.com	<1	Internet Data
21	A multivariate meta-analysis of student misbehavior and teacher burnou by Aloe-2014	<1	Publication
22	worldanimalsvoice.com	<1	Internet Data
23	docplayer.net	<1	Internet Data
24	ecologyandsociety.org	<1	Internet Data
25	en.wikipedia.org	<1	Internet Data
26	homedocbox.com	<1	Internet Data
27	ir.busitema.ac.ug	<1	Internet Data
28	link.springer.com	<1	Internet Data
29	Metric currents and Alberti representations by Schioppa-2016	<1	Publication
30	qdoc.tips	<1	Internet Data
31	scholarworks.umass.edu	<1	Publication

32	Theory of monolayers with boundaries Exact results and perturbative analysis by Rudnick-1999	<1	Publication
33	wcd.copernicus.org	<1	Internet Data
34	www.freepatentsonline.com	<1	Internet Data
35	www.readbag.com	<1	Internet Data
36	www.yumpu.com	<1	Internet Data